

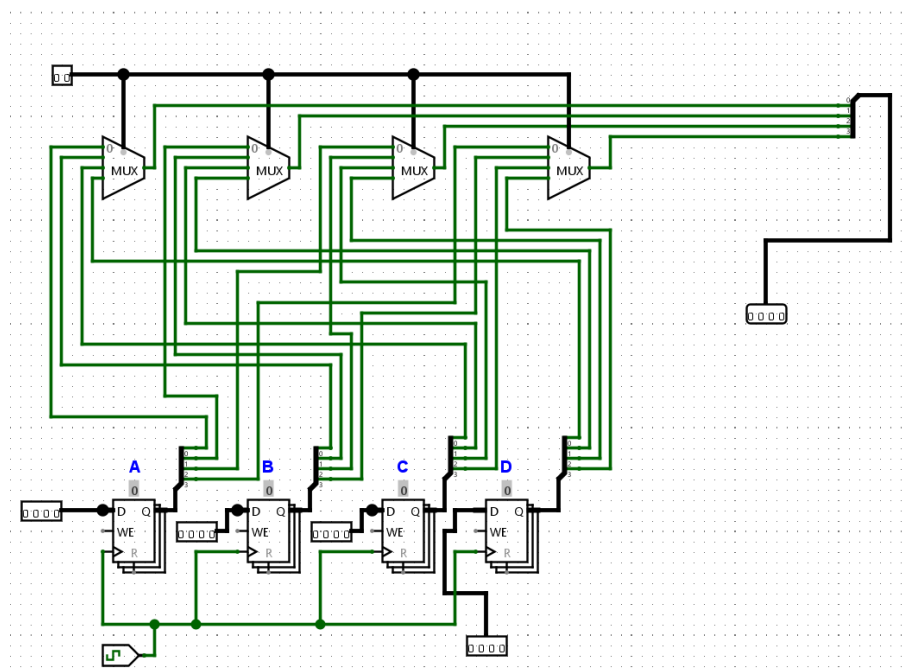
Lab 6: Implementation of 4-bit Common Bus System

Aim

To design and simulate a 4-bit Common Bus System using multiplexers and registers in Logisim Evolution.

Procedure

1. Open Logisim Evolution.
2. Create multiple 4-bit registers (e.g., R1, R2, R3, R4).
3. Use multiplexers to design a common bus system.
4. Connect outputs of all registers to the inputs of multiplexers.
5. Use select lines to choose which register places its data on the bus.
6. Connect the bus to the input of destination registers.
7. Add control signals for loading data into registers.
8. Use input switches to provide data and control signals.
9. Simulate the circuit by selecting different registers and observe data transfer through the bus.



Result

The 4-bit Common Bus System was successfully designed and simulated using Logisim Evolution.

- Data transfer between registers was achieved through a shared bus.
- Multiplexers correctly selected the desired register output.
- The system functioned as expected for different control inputs.

The experiment demonstrated the working of a common bus system in computer architecture.